



Role of digital finance, investment, and trade in technological progress

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ARTICLE INFO

JEL codes:

G10

F12

D92

C23

Keywords:

Digital finance

Technological progress

Trade openness

Investment

Panel data

ABSTRACT

The primary purpose of this study is to evaluate the impacts of digital finance, trade, and investment on technological progress in 30 developing Asian economies by employing the FMOLS (Fully Modified OLS) technique and annual data from 2000 to 2020. The findings concluded that digital finance has a positive impact on technology progress. Moreover, trade liberalization may accelerate technological progress. Foreign direct investment has no significant coefficient, interpreting that inward FDI (Foreign Direct Investment) to the examined developing Asian economies is unrelated to technological progress. The causality test confirmed a neutrality hypothesis for the causal relationship between FDI and technological progress. Furthermore, the bi-directional linkage between trade openness and technological progress is confirmed. The paper recommends that developing economies increase the share of investment in technological progress, prioritization of regionalism and trade multilateralism for technology transfer, and establishment of a “digital financial inclusion” framework.

1. Introduction

Technological progress is addressed as a way to upgrade the methods of producing commodities. The changes in technology in economic sectors can ensure a higher rate of economic growth, which is a want of countries in the post-COVID- era. Under the restrictions imposed by the governments to combat the outbreak of Coronavirus in 2020 and 2021, the economic sectors have experienced temporary and permanent stopping of activities, causing a deep recession worldwide. However, since 2022, due to vaccination and adaptation of people's lives to Corona conditions, it has been possible to return economic activities to the previous era. A new term under the title of “economic recovery” in the post-corona era has been formed by various scholars. Zhang and Dilanchiev (2022) express that economic recovery refers to the comeback of the economic sector after a severe recession during the pandemic. Nevertheless, Teng et al. (2022) and Zhao, Rasoulinezhad, Sarker, and Taghizadeh-Hesary (2023) mention the difficulties of economic recovery. They highlight the role of various drivers as critical engines of economic recovery pathways in countries.

Technology enhancement is one of the major critical success factors in realizing economic recovery in the short term. Generally, Solow's growth theory argues the importance of technological progress in increasing production efficiency and quality. From Solows' theory's perspective, long-term economic growth is only possible with technological advancement. In addition, with advanced technology, the costs of enterprises decrease over time, and consequently, higher profit is created. Gawal, Hulland, Kopalle, and Karahanna

Abbreviations: CTI, Composite Trade Index; DOLS, Dynamic OLS; FMOLS, Fully Modified OLS; SIT, Science, innovation and technology.

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<https://doi.org/10.1016/j.gfj.2023.100853>

Received 12 February 2023; Received in revised form 13 May 2023; Accepted 24 May 2023

Available online 29 May 2023

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(2020) and Taghizadeh-Hesary, Phoumin, and Rasoulinezhad (2023) declare that technology improvement profoundly affects firm profitability by lowering energy intensity and surging productivity. Even technological progress has a positive impact on human development. Perez- Castro, Mohamed-Maslouhi, and Montero- Alonso (2021) confirm that technological advancement can improve people's living conditions worldwide. It can lower income inequality, poverty, and environmental pollution as well.

In order to make technological progress, it is necessary to explore the influential factors on the progress. According to the assumption that technology is created inside or must be transferred from outside the country, three vital influential factors can be explained: trade, investment, and finance. By using the trade mechanism, it is possible to transfer new technologies from abroad. Through investment and financing, it will be possible to advance projects related to technological progress within a country. Maurer (2017) argues that trade liberalization allows a country to import technology and innovation from various trade partners. Malgouyres, Mayer, and Mazet- Sonilhac (2021) express that technological transfer from abroad can be upsized through foreign trade. Foreign trade can enhance the mobility of technology and innovation among trade partners (Chen & Wang, 2022). Another primary factor is an investment in projects relating to technological advancement. By increasing the volume of investment in the projects, the country's technological progress rate is improved. Innovative technologies may have a better diffusion through more investment in related projects. Bakhsh, Yin, and Shabir (2021) discuss that investment in technological progress helps countries deploy a more innovative technological base in production, transportation, and power generation, leading to economic welfare, environmental protection, and impressive profitability. However, it should be noted that investment in technology and innovation projects needs sufficient capital, which is a significant challenge for countries. Irandoust (2019) emphasizes the challenge of lack of capital promoting science, innovation, and technology (SIT) projects, especially in developing economies where the governments have a limited budget to invest and the participation of private sectors in investing is low. Haruvy, Katok, Ma, and Sethi (2019), Zakari and Khan (2022), and Afzal, Rasoulinezhad, and Malik (2022) highlight the necessity of financing tools to increase the attraction points of investment in technological progress for private investors and investors abroad.

The finance of technology and innovation projects means using different financial tools to attract capital, reduce risk, and provide transparent information related to technological advancement projects. Financing projects through the digital platform is a new method of finance that can remove time and geographical limitations and make the information and mechanism of projects related to technological progress more transparent. Al-Smadi (2022) confirms that digital finance directly and positively impacts financial inclusion in different economic sectors. He, Li, Geng, and Wen (2022) believe that digital finance accelerates the speed of the investment process in different projects and makes the investment market more transparent and structured.

The primary purpose of this study is to evaluate the impacts of digital finance, trade, and investment on technological progress in 30 developing Asian economies. The reasons for choosing the developing Asian countries are summarized into "the challenge of capital in these economies" and "necessary to advance technologies." Zhang (2022), Zakari (2022), and Zakari, Oryani, Alvarado, and Mumini (2023) argue that developing Asian economies like China do not have sufficient capital to promote all economic projects. Hence, efficient financing tools are necessary to help them to promote investment markets. According to the IMF report (IMF, 2022), the problem of investment in technology in Asian countries has been raised by geopolitical tensions and the US-China trade war.

Furthermore, despite successful economic growth experiences based on technological advances in some countries like Japan, most Asian developing countries need advancement in innovation and technology to reach adequate economic growth and development. Konstadakopolus (2002) expresses that technological development is a complex challenge in Asia and needs attention from scholars and policymakers. Woetzel and Seong (2021) note that Asian economies have enormous potential for technological progress, supported by over 52% of global growth in tech-company revenues in Asia during the last decade. However, the coronavirus's geopolitical tensions and consequences have made the region weaker in the field of technological progress. Hence, it is essential to explore the critical factors influencing technological advancement in Asia.

The contributions of this research to the existing literature are as follows: For the first time, a panel of 30 developing Asian economies is analyzed with the target of exploring of nexus between trade, investment, digital finance, and technological progress. The digital finance index is calculated using the Global Findex Database 2021 and ITU databases. In addition, The CTI (Composite trade index) is calculated for the examined Asian countries as a proxy for trade. Following Popova and Rasoulinezhad (2016), the composite trade index is more comprehensive than famous trade openness as the CTI addresses the relative country's trade flow to economic size and the contribution to global trade. The method of Nissan et al. (2012) is employed to calculate the technology diffusion index for the examined developing Asian economies.

The findings of the research concluded that digital finance has a positive impact on technology progress. Moreover, any improvement in trade liberalization may accelerate technological progress. Foreign direct investment has no significant coefficient, interpreting that inward FDI to the examined developing Asian economies is unrelated to technological progress. The causality test confirmed a neutrality hypothesis for the causal relationship between FDI and technological progress. Furthermore, the bi-directional linkage between trade openness and technological progress is confirmed. The paper recommends that developing economies increase the share of investment in technological progress, prioritization of regionalism and trade multilateralism for technology transfer, and establishment of a "digital financial inclusion" framework.

In order to complete the contributions, Section 2 argues the literature review. Section 3 provides the theoretical background. Following Section represents data information and model specification. Section 5 discusses empirical findings. The last Section summarizes the paper with conclusions, practical policies, and recommendations to research further.

2. Literature review

Technological progress has been considered a flourishing economic driver in the last decades. Farmer and Lafond (2016) expressed

that technological advancement is essential for economic growth and productivity. Jin, Duan, and Tang (2018) focused on technological progress's role in mitigating environmental pollution. They concluded that technological innovation could help countries lower fossil fuel consumption through energy efficiency progress and green energy deployment. Cansino, Roman- Collado, and Molina (2019) expressed that technological progress is necessary for creating a green economy and sustainable development, leading to better living conditions on Earth. Gaworski (2021) believed that technology is an element of development and that innovative technologies lead to higher economic productivity, profitability, and quality of commodities. Li, Zhou, and Yan (2022) addressed technological improvement as an efficient policy of governments to enhance social welfare and human development.

A group of scholars has emphasized the significant role of trade in technological progress. For example, Shabbir, Arshad, Alvi, and Iftikhar (2022) argued that trade policy uncertainty and trade openness impact the progress of innovation and technology in the medical products of developed countries. Employing the ARDL (Autoregressive Distributed Lag) model, they confirmed the negative and positive significant impacts of uncertainty and trade liberalization on technological progress, respectively. Mohamed, Liu, and Nie (2022) explored the relationship between technological innovation, economic growth, and trade openness. The significant findings depicted the positive linkage between trade openness and technological advancement. Craiut et al. (2022) noted that technological transfer from foreign countries could be developed through trade mechanisms. They mentioned that improving trade infrastructures and the financial support of SMEs by governments can make the technology transfer process more accessible.

Another strand of literature is concentrated on the role of investment in technological progress. Santos and Qin (2019) declared that many countries had prioritized investment in technology and innovation due to the role of technology in economic flourishing and social welfare. Zhang (2022) and Sun, Gao, Raza, Shah, and Sharif (2023) emphasized R&D investment as a factor determining technological progress. Sheng and Montgomery (2022) discussed that investment in R&D can increase technological diffusion in developing economies. By improving the state of research and development in countries through increasing public and private sector investment, the possibility of creating new technologies and commercializing these ideas will be created.

Digital finance is another influential factor in technological progress studied by a group of scholars. Tu, Rasoulinezhad, and Sarker (2020) discuss that digital finance profoundly impacts the expansion of green financing as the digital platform can boost accessibility and transparency in the market. In a study, Cao, Nie, Sun, Sun, and Taghizadeh-Hesary (2021) investigated China's digital finance. The research remarked on the positive role of digital finance on technological innovation. They argued that financial markets in the digital platform could improve the accessibility and reliability of services. Thathsarani and Jianguo (2022) argued that digital finance could lead to strengthened financial inclusion, causing improvements in technology and innovation. Al-Smadi (2022) contributed to the literature by studying the impacts of financial inclusion on digital finance in MENA (the Middle East and North Africa) countries. They revealed that financial inclusion and digital finance have interconnections, and expansion of technological deployment in financial markets scales up the access to credit facilities for firms and households. Mu, Liu, Tao, and Ye (2023) expressed that digital finance can improve corporate environmental, social, and governance (ESG) by providing higher quality-based services without geographical concerns. It can improve social equality throughout a country and helps the country to have a more efficient financial circle.

The gaps in the earlier literature are the lack of in-depth study about the technological progress in Asian developing economies, in addition to the absence of exploring the nexus between trade, investment, digital finance, and technological progress in the region of Asia. This research seeks to fill in these literature gaps.

3. Theoretical discussion

Theoretical discussion can be explored from two different aspects. First, the importance of technological progress in economic development and growth is essential to discuss. The second aspect explains the transmission channels through that digital finance can promote technological progress in a country.

Based on Solow's neoclassical theory of economic growth, technology, in addition to population and savings rate, is a crucial driver of the long-term GDP growth of countries. With technological progress, firms would gain more profits, leading to more production, employment, and GDP growth. Walheer (2021) believes technological progress is a chief element of Solow's exogenous growth theory and needs efficient governmental policies to enhance the local economy or be transferred from abroad. Therefore, technological progress is an important economic factor for countries. However, it is affected by various factors. The second aspect of the theoretical background determines the transmission channels of impacts of variables on technological progress. Shahebrahimi, Mahdad, Aliha, and ad Naeini, A. (2022) and Sun, Gao, Tian, & Guan, 2023, Sun, Gao, & Razzaq, 2023) declared that countries could import technology from trading partners called "technological transfer." Martinez- Zarzoso and Chelala (2021) expressed that through trade integration with countries, the technology transfers increase, leading to more productivity and growth in industries and other economic sectors. In other words, countries can trade advanced technologies among themselves, leading to the same distribution of technologies in different economic sectors. Investment is another critical factor that strengthens the ability to create innovation and new technologies within a country.

Regarding technological progress, investment can be implemented in R&D (Research & Development) or establishing an innovative technology project. Alam, Uddin, Yazdifar, Shafique, and Lartey (2020) discussed that R&D investment promotes a knowledge-based economy, and patents and innovations cause technological progress. Tran (2022) mentioned that countries should prioritize investment in R&D as an efficient policy to promote local innovative technological potentials. Gurtuev, Derkach, Makhosheva, and Ivanov (2020) and Sun, Guan, Mehmood, and Yang (2022) addressed the investment in innovation projects and confirmed that the investment could raise new technologies employed in different economic sectors. Finance and its digital type are other critical success factors of technological progress. According to Daud and Ahmad (2022), digital finance through financial inclusion promotes poverty eradication and access to financial services and helps commercialize technological progress projects. In another study, Wang, Tang, and Hu

(2022) declared that digital finance can lower financial services' costs and promote sustainable innovation and technology, leading to sustainable development worldwide.

4. Data and estimation strategy

In order to evaluate the impacts of trade, investment, and digital finance on technological progress, a group of 30 developing Asian economies (i.e., China, India, Indonesia, Malaysia, The Philippines, Thailand, Vietnam, Bangladesh, Cambodia, Mongolia, Myanmar, Nepal, Sri Lanka, Pakistan, Iran, Saudi Arabia, Uzbekistan, Kazakhstan, Jordan, Azerbaijan, Tajikistan, Turkmenistan, Kuwait, Armenia, Qatar, Bahrain, Bhutan, Maldives, Brunei, The United Arab Emirates) is chosen to analyze through the panel data approach. The access and availability of data on variables limited the number of countries to study. Regarding variables, the method of Nissan et al. (2012) is employed to calculate the technology diffusion index as the dependent variable. The index comprises telephone mainlines (per 1000 people), cellular subscribers (per 1000 people), internet users (per 1000 people), patents granted to residents (per million people), R&D expenditures (% of GDP), and researchers in R&D (per million people). The digital finance index as an explanatory variable is calculated using the Global Findex Database 2021 and ITU databases. The index includes using a mobile phone or the internet to pay bills (% age + 15), using a mobile phone or the internet to send money (% age + 15), using a mobile phone or the internet to buy something online (% age 15+), Saved money using a mobile money account (% age + 15), and owns a debit card (% age 15+). The CTI (Composite trade index) is calculated based on Popova and Rasoulnezhad (2016) for the examined Asian countries as a proxy for trade. Eq. 1 represents the formula for the calculation of the CTI as follows:

$$CTI_i = \frac{n(X + M)_i^2}{GDP_i \sum_{j=1}^n (X + M)_j} \quad (1)$$

Where the trade volume of country *i* shows by *X* + *M*, GDP denotes economic mass of country *i*, whereas the world trade is represented by $\sum_{j=1}^n (X + M)_j$.

Foreign direct investment is chosen as the following explanatory variable. In addition, the inflation rate is selected as the control variable. The impacts of the general price level of commodities on technological progress have been addressed by Evers, Stefan, and Schiffbauer (2020) and Poloz (2021). Table 1 represents the primary description of selected variables. The data of variables are gathered from 2000 to 2020.

Before estimating the coefficients of the variables, the stationary of the series is checked through Fisher-PP (Maddala & Wu, 1999) nonparametric test. The Levin, Lin, and Chu (LLC) and Im, Pesaran, and Shin (IPS) panel unit root tests are conducted as the robustness check for stationary checking of the series. Next, the panel co-integration tests are employed, proposed by Westerlund (2007) and Persyn & Westerlund, 2008. The Kao residual co-integration test is used as the robustness check for exploring the relationship between the series in the long term. The Fully Modified OLS (FMOLS) is conducted to estimate the coefficients of the examined variables. The primary advantages of the FMOLS technique are its ability to correct endogeneity bias and serial correlation problems. Moreover, the DOLS (Dynamic OLS) approach is employed as the robustness estimation check of the empirical results. Lastly, Dumitrescu and Hurlin (2012) pairwise causality approach explores the causality relationship between variables.

5. Empirical findings and discussion

Before conducting estimations of coefficients of independent variables, preliminary tests should be applied. First, the Fisher-type unit root test is employed to explore the stationary of the series in the panel. Table 2 reports the results of the panel unit root tests and concludes that all variables become stationary after the first difference, expressing that all series are I (Afzal et al., 2022).

Levin, Lin, and Chu (LLC) and Im, Pesaran, and Shin (IPS) panel unit root tests are employed as the robustness check for stationary checking of variables (Table 3). The robustness checks' findings align with the earlier result and confirm that the variables are stationary at first difference.

Since the variables are stationary at I (Afzal et al., 2022), the panel co-integration test of Westerlund (2007) can be carried out. Table 4 lists the results of the co-integration check and confirms that variables move together in the long term, meaning a co-integration relationship exists between variables.

In continuation, Kao's test is used to ascertain the reliability of Westerlund's co-integration test. According to Table 5, Kao's test supports the earlier findings and confirms the presence of a long-term co-integration relationship between variables.

Table 1
Variables' description and specifications.

Definition of variable	Symbol	Unit of measurement	Source of collection
Technology diffusion index	TDI	Index	Calculated based on Nissan et al. (2012)
Digital finance index	DFI	Index	The Global Findex Database 2021, and ITU database
Composite trade index	CTI	Index	Calculated based on Popova and Rasoulnezhad (2016)
Foreign direct investment	FDI	BoP, current US dollars	World Bank
Inflation rate	INF	Annual %	World Bank

Source: Authors.

Table 2
Panel unit root test.

Variable	ADF		Phillips- Perron	
	At level	At first difference	At level	At first difference
TDI	0.1572	−10.293**	0.0938	−14.399**
DFI	2.6943	−9.0832**	2.7931	−21.0739**
CTI	2.7811	−7.8449*	1.0383	−14.9093**
FDI	0.1830	−9.4820**	0.07882	−17.4934**
INF	3.1193	−10.0119	1.7664	−20.3880**

Note 1: TDI, DFI, CTI, FDI, and INF are technology diffusion index, digital finance index, composite trade index, foreign direct investment, and inflation rate, respectively.

Note 2: * and ** represent significance at 10% and 5%, respectively.

Source: Authors.

Table 3
Robustness check for stationary check.

Variable	Levin, Lin and Chu		Im, Pesaran and Shin W-stat	
	At level	At first difference	At level	At first difference
TDI	2.5498	−7.101**	3.4938	−8.3844*
DFI	−0.3394	−5.4977**	4.0944	−6.7758**
CTI	1.6640	−8.1194**	1.2294	−10.5672**
FDI	−1.2394	−8.3923**	1.2774	−8.5733**
INF	0.1194	−6.4495**	1.8845	−9.5223**

Note 1: TDI, DFI, CTI, FDI, and INF are technology diffusion index, digital finance index, composite trade index, foreign direct investment, ad inflation rate, respectively.

Note 2: * and ** represent significance at 10% and 5%, respectively.

Source: Authors.

Table 4
Panel co-integration test.

Statistic	Intercept and trend	Intercept
G_t	−2.8338**	−2.9091**
G_a	−9.838	−12.294
P_t	−11.093**	−13.1003*
P_a	−10.384**	−16.6129**

Note: * and ** represent significance at 10% and 5%, respectively.

Source: Authors.

The FMOLS estimation technique is used to address long-term coefficients of independent variables. Table 6 represents the findings of FMOLS estimation as follows:

According to the empirical estimations reported in Table 5, digital finance positively and significantly impacts technology progress in the examined developing Asian economies. A 1% improvement in digital finance leads to an increase in technological advancement by nearly 0.58%. The finding supports the earlier literature like Cao et al. (2021), Thathsarani and Jianguo (2022), and Mu et al. (2023), who confirmed the significant role of digital finance on innovation and technology improvement. Digital finance can lower the costs of financial services and promote sustainable innovation and technology, leading to sustainable development worldwide. Finance and its digital type is the critical success factor of technological progress. Digital finance through financial inclusion promotes poverty eradication and access to financial services and helps commercialize technological progress projects. Regarding trade openness, the

Table 5
Robustness check for the panel co-integration check.

–	t-stat	p-value
ADF	−2.5593**	0.039
Residual variance	0.0007083	–
HAC variance	0.0006193	–

Note 1: HAC variance stands for the Heteroscedasticity and autocorrelation consistent variance.

Note 2: ** shows significance at 5% level.

Source: Authors.

Table 6
FMOLS estimation results.

Independent variable	Coefficient	p-value
DFI	0.582	0.0031
CTI	0.381	0.0411
FDI	0.103	0.1839
INF	−0.0319	0.0077

Note: DFI, CTI, FDI, and INF are digital finance index, composite trade index, foreign direct investment, and inflation rate, respectively.

Source: Authors.

coefficient is positive and significant, expressing that any improvement in trade liberalization may accelerate technological progress in the examined developing Asian economies.

The finding is in line with [Shabbir et al. \(2022\)](#), [Mohamed et al. \(2022\)](#), and [Craiut et al. \(2022\)](#). By using the trade mechanism, it is possible to transfer new technologies from abroad. In addition, trade liberalization provides opportunities for a country to import technology and innovation from various trade partners. Trade openness can enhance the mobility of technology and innovation throughout the trade partners. Foreign direct investment has no significant coefficient, interpreting that inward FDI to the examined developing Asian economies is unrelated to technological progress. The main reason is that there are low-tech firms with unskilled workers in developed Asian countries that cannot use the opportunities of FDI to enhance technology and innovation in the production process. The finding is in contrast to [Santos and Qin \(2019\)](#), [Zhang \(2022\)](#), and [Sheng and Montgomery \(2022\)](#), who expressed that investment and technological progress have a positive linkage. Lastly, the inflation rate has a negative coefficient, meaning that a 1% increase in the general price level of commodities may reduce technological progress by approximately 0.03% in the selected developing Asian countries. The finding supports the earlier literature like [Evers et al. \(2020\)](#) and [Poloz \(2021\)](#), who declared that the inflation rate increases the costs of technological transfer from abroad and finding projects related to innovation and technological progress.

The DOLS estimation technique is conducted ([Table 7](#)) as the robustness check for the FMOLS findings. [Table 7](#) depicts the consistency in signs of coefficients of the independent variables, ascertaining the results of the FMOLS estimation approach.

The causality relationship through the Dumitrescu-Hurlin test is analyzed in the last stage. [Table 8](#) reports the results of the causality relationship between variables:

According to [Table 8](#), there is a bi-directional causal linkage between technological progress and trade openness, whereas there is no causal relationship between technological progress and FDI. Moreover, there is a uni-directional nexus between digital finance and technological progress. Hence, it can be concluded that there is a neutrality hypothesis for the causal relationship between FDI and technological progress in the examined developing Asian economies, highlighting the importance of FDI regulations reforming or leading FDI inwards into projects related to technological progress. Furthermore, the bi-directional linkage between trade openness and technological progress in the examined Asian countries reveals that all the two variables may be sensitive to any implemented policies in the trade sector or technological progress issues.

6. Conclusions and policy recommendations

Technological progress is addressed as a way to upgrade the methods of producing commodities. The changes in technology in economic sectors can ensure a higher rate of economic growth, which is a want of countries in the post-COVID- era. The primary purpose of this study is to evaluate the impacts of digital finance, trade, and investment on technological progress in 30 developing Asian economies by employing the FMOLS (Fully Modified Ordinary Least Squares) technique and annual data from 2000 to 2020. The research findings concluded that digital finance positively and significantly impacts technology progress in the examined developing Asian economies. Digital finance can lower the costs of financial services and promote sustainable innovation and technology, leading to sustainable development worldwide.

Regarding trade openness, the coefficient is positive and significant, expressing that any improvement in trade liberalization may accelerate technological progress in the examined developing Asian economies. In addition, trade liberalization provides opportunities for a country to import technology and innovation from various trade partners. Trade openness can enhance the mobility of technology and innovation throughout the trade partners. Foreign direct investment has no significant coefficient, interpreting that inward FDI to the examined developing Asian economies is unrelated to technological progress. The inflation rate has a negative coefficient, meaning that a 1% increase in the general price level of commodities may reduce technological progress by approximately 0.03% in the selected developing Asian countries. The causality test results confirmed a neutrality hypothesis for the causal relationship between FDI and technological progress in the examined developing Asian economies, highlighting the importance of FDI regulations reforming or leading FDI inwards into projects related to technological progress. Furthermore, the bi-directional linkage between trade openness and technological progress in the examined Asian countries reveals that all the two variables may be sensitive to any implemented policies in the trade sector or technological progress issues.

The summarization of concluding remarks opens the door to practical policies for developing Asian economies. Considering the lack of connection between investment and technological progress in these countries, it is suggested that governments plan to gradually increase the share of investment in technological progress compared to the total investment. Currently, developing countries

Table 7
Robustness test for ensuring the validation of FMOLS results.

Independent variable	Coefficient	p-value
DFI	0.048	0.0411
CTI	0.009	0.0087
FDI	0.011	0.2664
INF	−0.336	0.0189

Note: DFI, CTI, FDI, and INF are digital finance index, composite trade index, foreign direct investment, and inflation rate, respectively.

Source: Authors.

Table 8
Panel causality test.

H0	W statistic	Z bar statistic	p-value
TDI does not homogeneously cause DFI	5.483	8.239	0.004
DFI does not homogeneously cause TDI	1.382	−0.738	0.388
TDI does not homogeneously cause FDI	2.573	1.066	0.189
FDI does not homogeneously cause TDI	1.522	−0.818	0.577
TDI does not homogeneously cause CTI	4.442	5.693	0.018
CTI does not homogeneously cause TDI	3.452	2.018	0.005

Note: TDI, DFI, CTI, and FDI, are technology diffusion index, digital finance index, composite trade index, and foreign direct investment, respectively.
Source: Authors.

in Asia primarily depends on the transfer of technology from advanced countries, which will threaten the security of these countries' technology in the future. Therefore, relying on the internal power of innovation and technological advancement through investment will be a suitable practical policy for the developing countries of Asia. Prioritizing the policies of regionalism and trade multilateralism in the field of technology transfer and innovation is another practical policy for developing countries. Many free trade agreements in different regions of Asia can be focused on transferring knowledge, innovation, and technology (KIT). Another practical policy is the development of the digital financial market through de-bureaucracy, upgrading the level of the Internet network, increasing the level of people's access to the Internet and mobile network, transparency of regulations, and the establishment of a “digital financial inclusion” framework to combat financial poverty and increase social justice. Taking advantage of new financing markets such as green bonds is another crucial practical policy for developing countries in Asia. Due to financial problems and high investment risk in most countries, the private sector wants to refrain from participating in eco-friendly economic projects. Therefore, practical green financing tools such as green bonds can help attract more private sector funds for environmentally friendly projects such as green technologies in these countries. The support of governments in the developing countries of Asia for patenting, ideation, and commercialization of innovations is highly recommended in most knowledge-based economy programs because the mentioned practical policy will create new technologies from within the countries themselves and will reduce the dependence on developing Asian nations on western knowledge and industrialized countries.

The findings of this research can be used and developed by future researchers. It is suggested to study developed and developing countries and compare and interpret the results obtained from the relationship between technological progress, trade, investment, and digital finance in these two groups of countries. It is also suggested to use other technological progress indicators to compare the explanatory variables' effects in the econometric model. Considering the developments caused by the corona pandemic in recent years, it is suggested that corona be included as a control variable in the econometric model so that the relationship between the explanatory variables and the dependent variable can be estimated and interpreted in terms of the effect of the corona disease. A survey method (questionnaire or interview) with experts on the success factors in advancing technology in developing Asian countries is suggested for future research. Experts' opinions will provide more accurate practical tips to policymakers in developing countries in Asia.

Author statement

Conceptualization, methodology, data curation.

Funding details

This work was supported by The Key R&D and Promotion Project (Soft Science Research) in Henan Province, China (222400410057).

Declaration of Competing Interest

The authors declare that they have not known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

References

- Afzal, A., Rasoulinezhad, E., & Malik, Z. (2022). Green finance and sustainable development in Europe. *Economic Research- Ekonomska Istrazivanja*, 35(1), 5150–5163.
- Alam, A., Uddin, M., Yazdifar, H., Shafique, S., & Lartey, T. (2020). R&D investment, firm performance and moderating role of system and safeguard: Evidence from emerging markets. *Journal of Business Research*, 106, 94–105.
- Al-Smadi, M. (2022). *Examining the relationship between digital finance and financial inclusion: Evidence from MENA countries*. Borsa Istanbul Review. <https://doi.org/10.1016/j.bir.2022.11.016>. In Press.
- Bakhsh, S., Yin, H., & Shabir, M. (2021). Foreign investment and CO2 emissions: Do technological innovation and institutional quality matter? Evidence from system GMM approach. *Environmental Science and Pollution Research*, 28, 19424–19438.
- Cansino, J., Roman-Collado, R., & Molina, J. (2019). Quality of institutions, technological Progress, and pollution havens in Latin America. An analysis of the environmental Kuznets curve hypothesis. *Sustainability*, 11(13), 3708. <https://doi.org/10.3390/su11133708>
- Cao, S., Nie, L., Sun, H., Sun, W., & Taghizadeh-Hesary, F. (2021). Digital finance, green technological innovation and energy-environmental performance: Evidence from China's regional economies. *Journal of Cleaner Production*, 327, Article 129458. <https://doi.org/10.1016/j.jclepro.2021.129458>
- Chen, M., & Wang, H. (2022). Import technology sophistication and high-quality economic development: Evidence from city-level data of China. *Economic Research*, 35(1). <https://doi.org/10.1080/1331677X.2021.1956362>
- Craiu, L., Bungau, C., Bungau, T., Grava, C., Otrisal, P., & Radu, A. (2022). Technology transfer, sustainability, and development, worldwide and in Romania. *Sustainability*, 14(15728). <https://doi.org/10.3390/su142315728>
- Daud, S., & Ahmad, A. (2022). Financial inclusion, economic growth and the role of digital technology. *Finance Research Letters*. <https://doi.org/10.1016/j.frl.2022.103602>. In Press.
- Dumitrescu, E. I., & Hurlin, C. (2012). Testing for granger noncausality in heterogeneous panels. *Economic Modelling*, 29, 1450–1460.
- Evers, M., Stefan, N., & Schiffbauer, M. (2020). Inflation, liquidity and innovation. *European Economic Review*, 128, Article 103506. <https://doi.org/10.1016/j.euroecorev.2020.103506>
- Farmer, J., & Lafond, F. (2016). How predictable is technological progress? *Research Policy*, 45(3), 647–665.
- Gawal, D., Hulland, J., Kopalle, P., & Karahanna, E. (2020). The future of technology and marketing: A multidisciplinary perspective. *Journal of the Academy of Marketing Science*, 48, 1–8.
- Gaworski, M. (2021). Implementation of technical and technological Progress in dairy production. *Processes*, 9(12), 2103. <https://doi.org/10.3390/pr9122103>
- Gurtuev, A., Derkach, E., Makhosheva, S., & Ivanov, Z. (2020). A Bayesian approach to investment in innovation projects with the presence of fake innovators. *Heliyon*, 6(11), Article e05603. <https://doi.org/10.1016/j.heliyon.2020.e05603>
- Haruvy, E., Katok, E., Ma, Z., & Sethi, S. (2019). Relationship-specific investment and hold-up problems in supply chains: Theory and experiments. *Business Research*, 12, 45–74.
- He, C., Li, C., Geng, X., & Wen, Z. (2022). Research on the influence of digital finance on local economic development. *Procedia Computer Science*, 202, 385–389.
- IMF. (2022). Asia and the World Face Growing Risks From Economic Fragmentation.. URL: <https://www.imf.org/en/Blogs/Articles/2022/10/27/asia-and-the-world-face-growing-risks-from-economic-fragmentation> accessed 08.01.2023.
- Irlandoust, M. (2019). Saving and investment causality: Implications for financial integration in transition countries of Eastern Europe. *International Economics and Economic Policy*, 16, 397–416.
- Jin, L., Duan, K., & Tang, X. (2018). What is the relationship between technological innovation and energy consumption? Empirical analysis based on provincial panel data from China. *Sustainability*, 10(1), 145. <https://doi.org/10.3390/su10010145>
- Konstadakopolus, D. (2002). The challenge of technological development for ASEAN: Intraregional and international co-operation. *ASEAN Economic Bulletin*, 19(1), 100–110.
- Li, X., Zhou, X., & Yan, K. (2022). Technological progress for sustainable development : An empirical analysis from China. *Economic Analysis and Policy*, 76, 146–155.
- Maddala, G., & Wu, S. (1999). A comparative study of unit root tests with panel data and a new sample test. *Oxford Bulletin of Economics and Statistics*, 61, 631–652.
- Malgouyres, C., Mayer, T., & Mazet- Sonilhac, C. (2021). Technology- induced trade shocks? Evidence from broadband expansion in France. *Journal of International Economics*, 133, Article 103250. <https://doi.org/10.1016/j.jinteco.2021.103250>
- Martinez- Zarzoso, I., & Chelala, S. (2021). Trade agreements and international technology transfer. *Review of World Economics*, 157, 631–665.
- Maurer, M. (2017). Supply chain trade and technological transfer in the ASEAN+3 region. *China Economic Review*, 46, 277–289.
- Mohamed, M., Liu, P., & Nie, G. (2022). Causality between technological innovation and economic growth: Evidence from the economies of developing countries. *Sustainability*, 14(6). <https://doi.org/10.3390/su14063586>
- Mu, W., Liu, K., Tao, Y., & Ye, Y. (2023). Digital finance and corporate ESG. *Finance Research Letters*, 51, Article 103426. <https://doi.org/10.1016/j.frl.2022.103426>
- Perez- Castro, M., Mohamed-Maslouhi, M., & Montero- Alonso, M. (2021). The digital divide and its impact on the development of Mediterranean countries. *Technology in Society*, 64, Article 101452. <https://doi.org/10.1016/j.techsoc.2020.101452>
- Persyn, D., & Westerlund, J. (2008). Error correction based cointegration tests for panel data. *Stata Journal*, 8, 232–241.
- Poloz, S. (2021). Technological progress and monetary policy: Managing the fourth industrial revolution. *Journal of International Money and Finance*, 114, Article 102373. <https://doi.org/10.1016/j.jimonfin.2021.102373>
- Popova, L., & Rasoulinezhad, E. (2016). Have sanctions modified Iran's trade policy? An evidence of Asianization and De-Europeanization through the gravity model. *Economics*, 4(4), 24. <https://doi.org/10.3390/economics4040024>
- Santos, R., & Qin, L. (2019). Risk capital and emerging technologies: Innovation and investment patterns based on artificial intelligence patent data analysis. *Journal of Risk Financial Management*, 12(4), 189. <https://doi.org/10.3390/jrfm12040189>
- Shabbir, M., Arshad, M., Alvi, M., & Ifikhar, K. (2022). Impact of trade policy uncertainty and sustainable development on medical innovation for developed countries: an application of DID approach. *Sustainability*, 15(49). <https://doi.org/10.3390/su15010049>
- Shahebrahimi, S., Mahdad, M., Aliha, M., & ad Naeini, A.. (2022). Transfer of space Technologies in Iran: Drivers and constraints of success. *Space Policy*. <https://doi.org/10.1016/j.spacepol.2022.101518>. In Press.
- Sheng, D., & Montgomery, H. (2022). Should firms in emerging markets invest in R&D? Evidence from China's manufacturing sector. *Journal of Risk and Financial Management*, 15(517). <https://doi.org/10.3390/jrfm15110517>
- Sun, Y., Gao, P., Raza, S. A., Shah, N., & Sharif, A. (2023). The asymmetric effects of oil price shocks on the world food prices: Fresh evidence from quantile-on-quantile regression approach. *Energy*, 270, Article 126812.
- Sun, Y., Gao, P., & Razzaq, A. (2023). How does fiscal decentralization lead to renewable energy transition and a sustainable environment? Evidence from highly decentralized economies. *Renewable Energy*, 206, 1064–1074.
- Sun, Y., Gao, P., Tian, W., & Guan, W. (2023). Green innovation for resource efficiency and sustainability: Empirical analysis and policy. *Resources Policy*, 81, Article 103369.
- Sun, Y., Guan, W., Mehmood, U., & Yang, X. (2022). Asymmetric impacts of natural resources on ecological footprints: Exploring the role of economic growth, FDI and renewable energy in G-11 countries. *Resources Policy*, 79, Article 103026.
- Taghizadeh-Hesary, F., Phoumin, H., & Rasoulinezhad, E. (2023). Assessment of role of green bond in renewable energy resource development in Japan. *Resources Policy*, 80, Article 103272. <https://doi.org/10.1016/j.resourpol.2022.103272>

- Teng, B., Wang, S., Shi, Y., Sun, Y., Wang, W., Hu, W., & Shi, C. (2022). Economic recovery forecasts under impacts of COVID-19. *Economic Modelling*, 110, Article 105821. <https://doi.org/10.1016/j.econmod.2022.105821>
- Thathsarani, U., & Jianguo, W. (2022). Do digital finance and the technology acceptance model strengthen financial inclusion and SME performance? *Information*, 13(8), 390. <https://doi.org/10.3390/info13080390>
- Tran, Q. (2022). Democracy and corporate R&D investment. *Borsa Istanbul Review*, 22(3), 465–476.
- Tu, C., Rasoulinezhad, E., & Sarker, T. (2020). Investigating solutions for the development of a green bond market: Evidence from analytic hierarchy process. *Finance Research Letters*, 34, Article 101457. <https://doi.org/10.1016/j.frl.2020.101457>
- Walheer, B. (2021). Labor productivity and technology heterogeneity. *Journal of Macroeconomics*, 68, Article 103290. <https://doi.org/10.1016/j.jmacro.2021.103290>
- Wang, Q., Tang, K., & Hu, H. (2022). The impact of digital finance on green innovation: Evidence from provinces in China. *Innovation and Green Development*, 1(1), Article 100007. <https://doi.org/10.1016/j.igd.2022.100007>
- Westerlund, J. (2007). Testing for error correction in panel data. *Oxford Bulletin of Economics and Statistics*, 69, 709–748.
- Woetzel, J., & Seong, J. (2021). What is driving Asia's technological rise?. URL: <https://www.mckinsey.com/mgi/overview/in-the-news/what-is-driving-asias-technological-rise> accessed 05.02.2023.
- Zakari, A. (2022). The role of green finance in promoting sustainable economic and environmental development. *Studies of. Applied Economics*, 40(3). <https://doi.org/10.25115/eea.v40i3.5398>
- Zakari, A., & Khan, I. (2022). The introduction of green finance: A curse or a benefit to environmental sustainability? *Energy Research Letters*, 3(3). <https://doi.org/10.46557/001c.29977>
- Zakari, A., Oryani, B., Alvarado, R., & Mumini, K. (2023). Assessing the impact of green energy and finance on environmental performance in China and Japan. *Economic Change and Restructuring*, 56, 1185–1199.
- Zhang, C. (2022). China's emergence and development challenges that China faces in Central Asia. *Asian Review of Political Economy*, 1. <https://doi.org/10.1007/s44216-022-00005-7>
- Zhang, Y., & Dilanchiev, A. (2022). Economic recovery, industrial structure and natural resource utilization efficiency in China: Effect on green economic recovery. *Resources Policy*, 79, 102958. <https://doi.org/10.1016/j.resourpol.2022.102958>
- Zhao, L., Rasoulinezhad, E., Sarker, T., & Taghizadeh-Hesary, F. (2023). Effects of COVID-19 on global financial markets: Evidence from qualitative research for developed and developing economies. *The European Journal of Development Research*, 35, 148–166.